The Reference Is the Bug: Difficulty-Mismatched Non-Member Sets Break Membership-Inference Auditing of Unlearning

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Abstract

Membership-inference attacks (MIAs) are the standard instrument for auditing whether a model has truly unlearned specified data, and every MIA-based audit depends on a *non-member reference set* assumed to be exchangeable with the forget set in all respects except membership. **That property is required but not automatic — it depends on how the reference was constructed, and it is essentially never checked.** We give a cheap probe that checks it, and show it discriminates. Scoring a benchmark's forget and holdout sets with a model that was never trained on either isolates pure difficulty, since such a model has no membership signal to exploit. On TOFU, the most widely used LLM unlearning benchmark, the designated holdout **fails**: it is systematically *easier* than the forget set by $\approx$0.33 nats/token (AUC 0.332, p $<$ 1e-15). On MUSE-News the same probe at the same sample size **passes** (AUC 0.522, p = 0.59, not significant), so the probe distinguishes a sound reference from an unsound one rather than always reporting a gap. The two differ in construction, which gives a concrete rule: **partition one corpus, do not regenerate a reference later** — MUSE-News splits a single collection pass, whereas TOFU's holdout was regenerated separately, months after the original release. We then show the TOFU offset has three consequences for MIA-based auditing. First, it renders absolute-threshold auditing **uncalibratable**: the holdout-derived null distribution collapses to a degenerate point mass, admitting no valid decision threshold at any target false-alarm rate. Second, on a controlled synthetic retention gradient with unambiguous ground truth, the offset produces **missed detections** — a difficulty-matched auditor flags residual retention across a band where a holdout-based auditor following standard practice reports clean forgetting. Third, it corrupts raw-score MIAs (LOSS, ZLib, Min-K, Min-K++) while leaving reference-normalized MIAs intact. The offset-robust alternative — gold-referenced (relative) rules — is immune by common-mode cancellation, but is *unavailable on TOFU as shipped*: the released retain checkpoint is a byte-identical duplicate of the full finetuned model, and the released unlearned checkpoints are empty repositories. We provide a difficulty-matched reference construction that restores calibration, and release all experiments with GPU-free reproducibility from committed result arrays.

1. Introduction

Machine unlearning aims to remove the influence of specific training data from a trained model, motivated by privacy regulation and the right to erasure. A central, unresolved difficulty is *verification*: given a model that claims to have forgotten a set of data, how do we confirm it actually did? Membership-inference attacks are the field's default answer. An MIA assigns each sample a membership score, and an audit declares forgetting successful when the forget set becomes statistically indistinguishable from a *non-member reference* — canonically, a held-out set the model never trained on.

The entire construction rests on an unstated assumption: that the non-member reference is *exchangeable* with the forget set in everything except membership. If the reference is intrinsically easier or harder than the forget set for reasons unrelated to training, the audit's calibration is corrupted at the source — before any attack, any threshold, any model. To our knowledge, no prior work checks whether this assumption holds on the benchmarks the field actually uses.

We check it. On TOFU [1] — a synthetic-author QA benchmark and the most cited LLM unlearning testbed — it fails: the designated holdout set is measurably easier than the forget set for a model that has seen neither. Because the probe model has no membership information, the gap is *pure difficulty*, not leakage. On MUSE-News [2], at the same sample size, the same probe finds nothing (§5.1) — so this is a diagnostic that discriminates, not one that fires everywhere, and the property it tests is one benchmarks can and do get right. We then trace what the TOFU offset does to MIA-based auditing:

- It makes absolute-threshold auditing **uncalibratable** (§6.1): the holdout-based null is a degenerate point mass with no threshold at any false-alarm rate.
- It causes **missed detections** (§6.2): on a controlled retention gradient, standard-practice holdout auditing reports "forgotten" where a difficulty-matched auditor correctly detects residual retention.
- It **corrupts raw-score MIAs but not reference-normalized ones** (§6.3), which lets us say precisely which of the field's shipped attacks inherit the bias.

We are careful about scope (§7). The failure is specific to *absolute-threshold* auditing — the "MIA at chance ⇒ forgotten" convention that is standard practice. Gold-referenced *relative* rules are immune, because the difficulty offset is common-mode across candidate and reference and cancels under subtraction. But the offset-robust mode is *unavailable on TOFU as shipped*: there is no valid gold retrain to reference, because the released retain checkpoint duplicates the full finetuned model and the released unlearned checkpoints are empty. The offset breaks the auditing mode the benchmark supports, and the mode that would survive it is the one the benchmark cannot supply.

Our contributions:

- **C1.** A measured, provenance-verified difficulty offset in TOFU's holdout set, localized to the holdout (not the forget set) and shown not to be a length artifact (§5) — together with a **positive control** on MUSE-News, where the identical probe at the identical sample size returns chance, establishing that the check discriminates and yielding the construction rule *partition one corpus, do not regenerate a reference later* (§5.1).
- **C2.** A demonstration that the offset makes absolute-threshold auditing uncalibratable, and a difficulty-matched reference that restores calibration (§6.1, §8).
- **C3.** A controlled demonstration that the offset causes missed detections, with a same-margin control isolating the cause to the reference set (§6.2).
- **C4.** A map of which MIAs inherit the bias — raw-score corrupted, reference-normalized intact — tied to the specific attacks shipped by the standard evaluation framework (§6.3).
- **C5.** A difficulty-matched reference construction and a fully GPU-free-reproducible artifact release (§8).

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2. Background and Preliminaries

This section fixes notation and states the audit procedure we analyse. Nothing here is novel; it is written out because the failure we identify lives in an assumption that is usually left implicit.

2.1 Unlearning

Let $\mathcal{D}$ be a training corpus and let a model θ_{full} be obtained by training on $\mathcal{D}$. A *forget request* designates a subset $F \subset \mathcal{D}$ to be removed; the complement $R = \mathcal{D} \setminus F$ is the *retain set*. An unlearning algorithm is a map

$$\mathcal{U} : (\theta_{\text{full}}, F, R) \mapsto \theta_{\text{un}}$$

whose output should behave as though F had never been in the training corpus. The reference point for "as though" is the *gold retrain* θ_{retrain}, trained from scratch on R alone. Exact retraining is the definition of success but is usually too expensive to run, which is precisely why approximate unlearning methods — and the auditing problem — exist.

Two further ingredients matter for auditing. A *holdout* set H consists of samples drawn from the same distribution as $\mathcal{D}$ but never included in training, and therefore never seen by θ_{full}, θ_{un}, or θ_{retrain}. The triple (F, R, H) — forget, retain, holdout — is the standard evaluation scaffolding shipped by unlearning benchmarks [1, 2].

2.2 Membership inference

A membership-inference attack equips a model with a real-valued *membership score* $s(x; \theta)$ intended to be systematically lower (more member-like) on training members than on non-members. The canonical black-box instantiation thresholds the model's loss on the sample [3]; variants normalise by a compression estimate of the sample's intrinsic complexity [4], aggregate only the least-likely tokens [5, 6], or calibrate against a reference model's score on the same sample [7].

Rather than fix a threshold, an audit measures *separability.* Given member scores $\{s(x)\}_{x \in F}$ and non-member scores $\{s(x)\}_{x \in H}$, the attack's power is the area under the ROC curve of $-s$ against the membership label,

$$\text{AUC} = \Pr\left[\, s(x_F) < s(x_H) \,\right] + \tfrac{1}{2} \Pr\left[\, s(x_F) = s(x_H) \,\right], \qquad x_F \sim F,\ x_H \sim H.$$

$\text{AUC} = 0.5$ means the two sets are indistinguishable to the attack.

2.3 The audit and the retention score

An audit converts separability into a decision. We use the retention score

$$\rho \;=\; \operatorname{clip}(2(\text{AUC} - 0.5),\, 0,\, 1) \in [0, 1],$$

so $\rho = 0$ when the forget set is indistinguishable from the reference and $\rho = 1$ under perfect separability. The clipping encodes an asymmetry: $\text{AUC} < 0.5$ means members look *less* member-like than non-members, which is not evidence of retention, so it floors at zero rather than being read as a negative result.

Because ρ is a statistic of finite samples, a raw value cannot be read as a verdict. The audit therefore calibrates: it constructs a *null distribution* $\mathcal{N}$ of ρ under the truly-forgotten condition, sets the decision threshold τ at the $(1 - \text{FAR})$ quantile of $\mathcal{N}$ for a target false-alarm rate, and reports

$$\texttt{RETENTION DETECTED} \iff \rho > \tau.$$

The comparison is strict: a candidate is flagged only when it *exceeds* the forgotten baseline, not when it merely equals it. In our setting $\mathcal{N}$ is obtained by bootstrap-resampling the member and non-member scores of a model known to be free of the forget set, and the realised false-alarm rate is reported as $\text{FAR} = \Pr_{\mathcal{N}}[\rho > \tau]$ rather than assumed.

2.4 The exchangeability assumption

Everything above depends on a premise that is rarely stated. Write the score of a sample as a difficulty term plus a membership term,

$$s(x;\theta) \;=\; d(x) \;+\; m(x;\theta),$$

where d captures how hard x is intrinsically — for a language model, how predictable its tokens are to *any* model of that class — and m captures the reduction attributable to having trained on x. An MIA audit attributes the entire F-versus-H score gap to m. That attribution is valid only if the difficulty terms are matched in distribution:

$$d(x_F) \;\overset{d}{=}\; d(x_H), \qquad x_F \sim F,\; x_H \sim H. \tag{$\star$}$$

We call ($\star$) the **exchangeability assumption**. If it fails — if the reference is intrinsically easier or harder than the forget set — then the measured AUC confounds difficulty with membership, and every quantity built on top of it (the retention score, the null, the threshold, the verdict) inherits the confound. No amount of sophistication in the *estimator* repairs a violation of ($\star$), because the violation is in the estimator's input.

The contribution of this paper is to observe that ($\star$) is directly testable at negligible cost, to test it on the benchmarks the field actually uses, and to trace the consequences where it fails.

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3. Related Work

Benchmarks: TOFU. TOFU [1] is the most-cited testbed for LLM unlearning. It consists of synthetic question–answer pairs about 200 fictitious authors, generated so that the content cannot appear in any pretraining corpus — an elegant design that isolates the effect of fine-tuning. The benchmark ships forget/retain partitions at several fractions (`forget01/05/10` with matched `retain99/95/90`) together with finetuned target models. Its headline finding is that no baseline unlearning method it evaluated achieved effective forgetting. Our work does not dispute any TOFU result; it examines a component added to the benchmark later — the `holdout` splits that serve as the non-member reference for MIA-based evaluation — and shows that this component violates ($\star$).

Benchmarks: MUSE. MUSE [2] evaluates unlearning along six axes, including verbatim memorisation, knowledge memorisation, privacy leakage, utility preservation, scalability, and sustainability. Its two corpora are news articles and books. Relevant here is its construction discipline: for the news corpus it collects a single pool of BBC articles published after August 2023 — chosen so the target model's pretraining could not have seen them — and partitions that one pool into disjoint forget, retain, and holdout subsets. We use MUSE-News as a positive control and find its reference satisfies ($\star$); §5.1 argues the difference from TOFU is exactly this construction discipline.

Meta-evaluation frameworks. A recent line of work standardises unlearning evaluation, asking whether the *metrics themselves* are faithful and robust, and consolidates many algorithms and metrics behind one interface. This work is closest in spirit to ours and supplies the concrete MIA implementations we analyse in §6.3; it also states that MIA-based privacy metrics presuppose clean i.i.d. holdouts or oracle retrain models. Our finding sits one level upstream: the presupposed clean holdout is itself miscalibrated, so the miscalibration propagates into the shipped metrics rather than being screened off by them.

Critiques of the MIA paradigm. A parallel line argues that a *failed* MIA does not license the conclusion that forgetting occurred, since unlearned samples occupy a feature-space region distinct from genuine non-members, and proposes training-free estimators of the non-member mixture proportion instead. That critique targets the *estimator*: it accepts a clean reference and argues the inference drawn from it is unsound. Ours is orthogonal and prior in the pipeline — the reference itself carries a difficulty offset, which corrupts any estimator built on it. The two compose rather than compete: a difficulty-matched reference is a precondition for the estimator-level fix to mean anything.

Membership-inference attacks. The loss-threshold attack [3] established the overfitting–leakage link and remains the canonical black-box baseline. Later work sharpened the statistic, not the reference: zlib normalisation divides loss by a compression estimate, targeting cheap generic text [4]; Min-K% scores only the least-likely tokens, on the premise that memorised text lacks low-probability outliers [5]; Min-K%++ adds a theoretically motivated next-token normalisation [6]. All four are *raw-score* attacks — they threshold an absolute score against a non-member reference, and so inherit any violation of ($\star$) directly. The reference-model line instead calibrates per-example against a second model's score on the *same* sample [7], which is what makes it structurally immune to a common-mode offset (§6.3).

Relative (gold-referenced) rules. Rather than comparing a candidate to an absolute chance level, some evaluations compare it to a gold retrain model measured on the same reference set [1, 2]. We show in §7 that these rules are robust to a difficulty offset by common-mode cancellation — and that this escape hatch is nonetheless unavailable on TOFU as shipped, because no valid gold retrain is actually distributed.

Positioning in one line each. Against meta-evaluation frameworks: they evaluate metrics against a reference; we show the reference is the problem. Against paradigm critiques: they fix the estimator; we show the input to the estimator is miscalibrated. Against relative rules: they are offset-robust but unavailable on the benchmark as shipped.

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4. Experimental Setup

Every experiment in this paper is a scoring pass: no model is trained, and no gradient is taken. This section specifies the data, models, and procedures precisely enough to reproduce each number.

4.1 Datasets and splits

TOFU is loaded from the canonical dataset repository, whose configurations are single `train` splits of `{question, answer}` string pairs. We use `forget10` (400 pairs, the 10% forget partition), `retain90` (3600 pairs, its complement), and `holdout10` (400 pairs, the designated non-member reference). We verified that `forget10` and `retain90` are disjoint and that their union is exactly the full 4000-pair corpus the target model was finetuned on, and that `holdout10` is disjoint from that corpus. Where a control requires a size-matched retain sample we take the first 400 pairs of `retain90`.

MUSE-News is loaded from its benchmark repository under the `privleak` configuration, which supplies `forget`, `holdout`, and `retain` as three splits of 100 raw-text passages each. The three are near-identical in length (8053 / 8051 / 8070 mean characters), which is a consequence of their being partitions of a single collection pass rather than independently generated sets.

Provenance. The holdout we test is the configuration that the standard evaluation framework's MIA configuration points to as its non-member set — not a deprecated split, but the current field-standard reference, added to the benchmark by its maintainers as the improved hold-out. (TOFU_MIA.yaml → TOFU_QA_holdout → locuslab/TOFU config holdout10; added 2025-03-27)

4.2 Models

The **target** model is the publicly released TOFU-finetuned Phi-1.5 checkpoint: a 1.3B-parameter decoder-only transformer [10] finetuned on the full 4000-pair TOFU corpus, so all of `forget10` and `retain90` are training members for it. It is distributed in bfloat16 across 341 parameter tensors.

The **probe / null** model is the corresponding *base* Phi-1.5 [10], never finetuned on TOFU and never trained on MUSE-News. Its role is central: having seen neither the forget set nor the reference set, its membership term m is identically zero on both, so any score gap it exhibits is attributable to the difficulty term d alone. It is a measuring instrument for ($\star$), not a model under audit. For MUSE-News the same base model serves as probe; it predates the post-August-2023 collection window, so it is blind to all three splits, and any residual overlap would fall on forget and holdout symmetrically and so could not manufacture a gap.

A practical note: the finetuned repositories ship no tokenizer files, and loading one from them silently yields a broken vocabulary rather than raising — the tokenizer must be loaded from the base model.

4.3 The difficulty score: answer-only, length-normalised NLL

All scores are per-token negative log-likelihood over the *answer span only*, under a prompt template identical for members and non-members. For a pair (q, a):

1. Form the prompt $P(q)$ from a fixed template and tokenize it; let ℓ be its token count.
2. Tokenize $P(q) \| a$ to obtain the full sequence $x_1, \dots, x_T$.
3. Run one forward pass and read the next-token log-probabilities $\log p_\theta(x_t \mid x_{<t})$.
4. Report

$$s(q, a; \theta) \;=\; -\frac{1}{T-\ell} \sum_{t=\ell+1}^{T} \log p_\theta(x_t \mid x_{<t}).$$

Two properties of this definition matter. It is **answer-only**: the question tokens are context and are not scored, so a question that is easy or hard to predict cannot move the score. It is **length-normalised**: dividing by the answer's token count means a long answer is not automatically scored as more surprising than a short one. Both choices remove obvious confounds that would otherwise be mistaken for the effect we are measuring.

MUSE-News is prose rather than question–answer pairs, so the analogous span is the whole passage. We obtain this through the identical code path by setting the prompt to the empty string, which reduces the definition above to a full-text length-normalised NLL, and we truncate every passage to a uniform 512-token prefix — the splits are already length-matched, and the probe model's context is 2048 tokens.

Separability is computed with the same ROC-AUC routine in every experiment, so numbers from different sections are directly comparable by construction.

4.4 The difficulty-matched reference

Given base-model scores on the forget and holdout sets, we construct a matched reference by 1:1 nearest-neighbour matching without replacement under a caliper:

1. Sort the forget examples by base-model NLL (deterministic order).

2. For each forget example in turn, find the unmatched holdout example whose base-model NLL is closest.
3. Accept the pair if the absolute difference is within the caliper $c = 0.05$ nats/token; otherwise leave the forget example unmatched.

Selection uses **only the probe model's scores** — never the target model's — so no membership information can leak into the construction of the reference. The result is reported with its cost: matching discards forget examples whose difficulty has no counterpart in the holdout, so the matched audit evaluates a difficulty-matched *subpopulation* of the forget set, and we report the residual imbalance alongside every matched number.

4.5 The synthetic retention gradient

> **This gradient is a synthetic probe, not an unlearning method.** It exists solely to supply graded retention with known ground truth. It must not be compared to published unlearning numbers. Real-method validation is deferred (§10).

To study graded auditing we require models with known, varying amounts of residual retention. The benchmark's released unlearned checkpoints are unavailable (§7), so we construct a controlled gradient by linear interpolation in weight space between the finetuned target and the base model,

$$\theta(\alpha) \;=\; (1-\alpha)\,\theta_{\text{baseline}} \;+\; \alpha\,\theta_{\text{base}}, \qquad \alpha \in [0,1],$$

evaluated at $\alpha \in \{0, 0.15, 0.3, 0.5, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1\}$. Ground truth is unambiguous by construction: every $\alpha < 1$ retains a non-zero share of the finetuned weights, so residual retention is present and a correct auditor must detect it.

The construction is gated rather than trusted. We assert that the two checkpoints share all 341 parameter tensors by name and shape before blending, and we require the endpoints to reproduce the models they interpolate: $\alpha = 0$ must recover the finetuned member NLL and $\alpha = 1$ must recover the base model's. Both hold — 0.084 against a logged 0.090, and 2.072 with maximum per-example drift 0.0000 against the independently computed base-model scores (341/341 state-dict tensors aligned). Had either endpoint missed, the middle of the ladder would not have been read at all.

4.6 Hardware and reproducibility

All scoring runs on a single consumer GPU (8 GB), in half precision, with batch size one; the full set of experiments is a few tens of minutes of compute. Every per-example score is committed as a `.npz` array alongside the code, and every table and figure in this paper is regenerated from those arrays with no GPU and no model download. The figure-generation and verification scripts are part of the released artifact.

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5. A Difficulty Offset in the Standard Reference (EXP-1)

Probe design. To isolate difficulty from membership, we score TOFU's forget and holdout sets with a model that was never trained on *any* TOFU data (base Phi-1.5 [10]). Such a model cannot exploit membership; any systematic score gap between forget and holdout is therefore attributable to intrinsic difficulty alone. We use answer-only, length-normalized negative log-likelihood (NLL) per token as the difficulty measure, and report separability as ROC-AUC between forget and holdout NLLs.

Result. Table 1 reports the probe. Against the never-trained model, forget10 and holdout10 separate at AUC 0.332 — far from the 0.5 that exchangeable sets would yield. The two controls localize the defect: forget10 vs retain90, two splits from the *original* TOFU generation run, separate at AUC 0.514 (matched to noise), while retain90 vs holdout10 separates at 0.310. The holdout is ≈0.33 nats/token easier than *both* original splits. The defect is in the holdout, not the forget set.

Not a length artifact. Scoring is length-normalized, and holdout answers are on average *longer* than forget answers (41.8 vs 36.1 mean tokens), which would bias separability in the opposite direction. The offset persists against this bias.

Table 1: Difficulty probe on TOFU under a never-trained model (base Phi-1.5), answer-only length-normalised NLL per token. A reference satisfying (⋆) would give AUC 0.5.

comparison	n	member NLL	non-member NLL	gap	AUC
forget10 vs holdout10	400 / 400	2.072	1.748	−0.324	**0.332**
control: forget10 vs retain90	400 / 400	2.072	2.106	+0.034	0.514
control: retain90 vs holdout10	400 / 400	2.106	1.748	−0.357	0.310

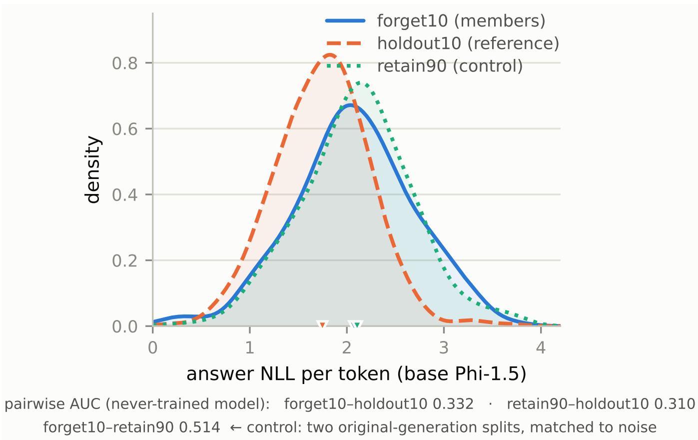

Figure 1: NLL distributions for forget10, holdout10, and retain90 under the never-trained model. The holdout distribution is visibly shifted toward lower NLL (easier); forget10 and retain90 nearly coincide, which is the control result.

5.1 Positive control: the probe passes a soundly-constructed reference (EXP-3)

A probe that reports a gap on every benchmark would be worthless — the TOFU result would say more about our difficulty measure than about TOFU. We therefore ran the identical probe, through the identical code path, on a second benchmark whose reference is built differently.

MUSE-News [2] (`muse-bench/MUSE-News`, config `privleak`) supplies `forget`, `holdout` and `retain` as three mutually disjoint subsets of a *single* corpus — BBC articles published after August 2023, collected in one pass so that the target model [11] had never seen them. Its holdout is documented as data never seen during pre-training or unlearning. We probe it with the same never-trained model (base Phi-1.5), which predates that collection window and is therefore blind to all three splits; any residual overlap would affect forget and holdout symmetrically and so could not manufacture a gap.

The probe passes. forget vs holdout gives **AUC 0.522 (p = 0.59, not significant)**, an offset of +0.039 nats/token — an order of magnitude smaller than TOFU's and in the opposite direction. Controls agree: forget vs retain 0.499 (p = 0.98), retain vs holdout 0.522 (p = 0.59). Every MUSE-News comparison is statistically indistinguishable from chance.

This is not a power artifact, and the matched-sample-size comparison is what shows it. MUSE-News supplies 100 texts per split. Restricting TOFU to the *same* n = 100 still yields **AUC 0.294 (p = 4.8e-07)** — far from chance. Same probe, same code, same sample size: one benchmark fails decisively, the other passes cleanly. The diagnostic discriminates.

What separates them is construction, not subject matter. MUSE-News partitions one corpus collected in a single pass; TOFU's holdout was regenerated separately, roughly fourteen months after the original release, and did not reproduce the original generation's difficulty distribution. Hence the rule we propose: *partition one corpus; do not regenerate a reference later.* Where regeneration is unavoidable, this probe is a cheap check — one forward pass per example with an off-the-shelf never-trained model, no training and no target-model access.

Table 2: The same probe on both benchmarks. The matched-sample-size row is the one that rules out low power as an explanation for the MUSE-News null.

benchmark	comparison	n	AUC	p	reference verdict
TOFU	forget10 vs holdout10	400 / 400	**0.332**	< 1e-15	violates (⋆)
TOFU	forget10 vs holdout10	100 / 100	**0.294**	4.8e-07	violates (⋆)
TOFU	*control:* forget10 vs retain90	400 / 400	0.514	n.s.	consistent with (⋆)
MUSE-News	forget vs holdout	100 / 100	**0.522**	0.59	consistent with (⋆)
MUSE-News	*control:* forget vs retain	100 / 100	0.499	0.98	consistent with (⋆)
MUSE-News	*control:* retain vs holdout	100 / 100	0.522	0.59	consistent with (⋆)

Excluded: MUSE-Books. **Its forget–holdout AUC of 0.302 superficially resembles TOFU's 0.332 and must not be read as corroboration** — the controls show its splits are incoherent: retain–holdout is 0.153 and forget–retain is 0.733, i.e. the three splits disagree with one another in a way no membership account explains. The cause is contamination: its text is Harry Potter, which essentially every general-purpose base LM has seen in pretraining, so no never-trained probe exists for it and any gap is uninterpretable — memorization cannot be separated from difficulty. Its numbers confirm that diagnosis, not our finding. We report it for completeness and rest nothing on it.

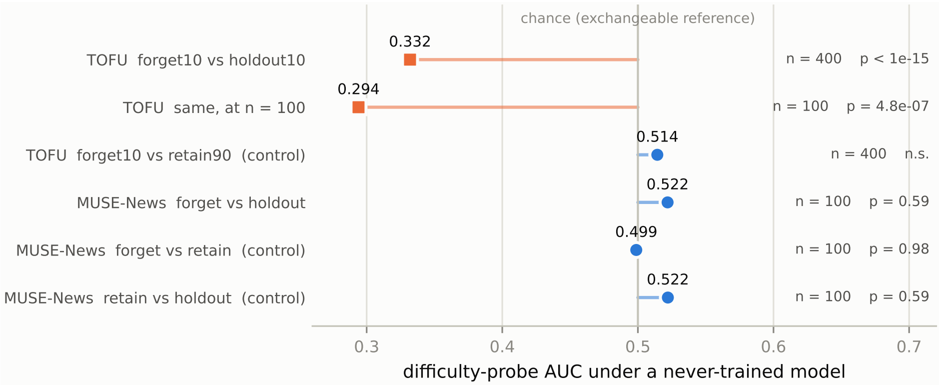

Figure 2: The probe discriminates. Identical instrument, identical code path: TOFU's reference fails at both sample sizes while every MUSE-News comparison sits at chance. The n = 100 TOFU row rules out low power as the explanation for the MUSE-News null.

6. Consequences for Auditing (EXP-2, EXP-2b)

We now trace what the offset does to a graded audit, using the synthetic retention gradient of §4.5. Table 3 gives the full ladder under all three scorings; the subsections take the three consequences in turn.

Table 3: The full retention ladder. *Naive acc.* is teacher-forced answer accuracy on the forget set — the quantity a naive evaluation would read. Retention is reported under the field-default holdout reference, under the difficulty-matched reference, and under a reference-normalised (per-example) scoring. The ladder is a **synthetic probe, not an unlearning method**.

α	naive acc.	ρ vs holdout10	ρ vs matched	ρ reference-norm.
0.00	0.989	0.997	0.996	0.999
0.15	0.964	0.993	0.991	0.997
0.30	0.875	0.973	0.981	0.994
0.50	0.731	0.733	0.878	0.985
0.70	0.640	**0.162**	**0.541**	**0.970**
0.75	0.622	0.040	0.439	—
0.80	0.608	0.000	0.339	—
0.85	0.594	0.000	0.245	—
0.90	0.582	0.000	0.160	—
0.95	0.573	0.000	0.081	—
1.00	0.563	0.000	0.019	0.000

6.1 The offset makes absolute-threshold auditing uncalibratable (C2)

An absolute-threshold auditor calibrates a decision threshold by bootstrapping a null distribution of the retention score under truly-forgotten conditions, then flags retention above the (1−FAR) quantile. Using holdout10 as the non-member reference, this null collapses to a **degenerate point mass** (standard deviation 0.0000), because the difficulty offset drives the truly-forgotten score identically to a floor. No valid threshold exists at any target false-alarm rate.

Replacing holdout10 with a difficulty-matched reference (§8) restores a well-defined null (standard deviation 0.0358), a usable threshold (0.0911), and the target false-alarm rate (0.050). Difficulty matching is what makes the auditor calibratable at all.

Table 4: Calibration under the two references. The holdout-derived null has no spread, so no quantile exists to threshold at — the auditor cannot be calibrated, at any target FAR.

reference	null s.d.	threshold τ	realised FAR	calibratable?
holdout10 (field default)	0.0000	0	—	**no** — point mass
difficulty-matched	0.0358	0.0911	0.050	yes

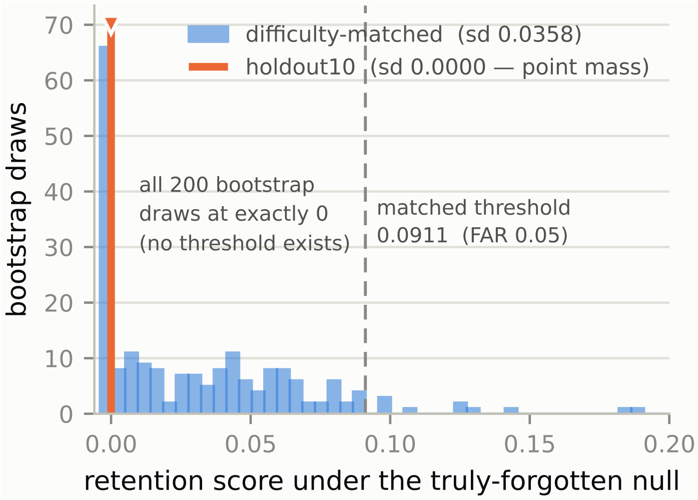

Figure 3: The two calibration nulls. Every bootstrap draw from the holdout-referenced null lands at exactly zero, so no quantile exists to place a threshold at; the difficulty-matched null has genuine spread and supports a threshold at the target false-alarm rate.

6.2 The offset causes missed detections (C3)

Figure 2 plots residual-retention score against α under both references, with decision thresholds. The flip is **directly observed at the sampled rungs** $\alpha \in$ **{0.80, 0.85, 0.90}**; taking linear crossings between rungs, the **missed-detection band spans** $\alpha \approx$ **0.77–0.94** (the holdout-based auditor stops detecting at $\alpha \approx 0.77$, the difficulty-matched auditor at $\alpha \approx 0.94$). Across this band, the difficulty-matched auditor reports RETENTION DETECTED while a holdout-based auditor following the standard “AUC $\approx$ 0.5 $\Rightarrow$ forgotten” convention reports clean forgetting. Ground truth is unambiguous (retention is present for all $\alpha < 1$), so these are genuine missed detections induced by the reference, not by chance.

Same-margin control (isolating the cause). One might object that the matched auditor is calibrated while the holdout auditor is not, confounding reference choice with calibration. We control for this by transplanting the matched threshold onto holdout-referenced scores: same decision margin, only the reference set differs. The flip persists (4/5 rungs in the fine sweep), isolating the cause to the **reference set**, not the calibration. The offset costs approximately one-fifth of the detectable retention range (holdout boundary $\alpha{\approx}0.77$ vs matched boundary $\alpha{\approx}0.94$).

6.3 Which MIAs inherit the bias (C4)

The offset corrupts *raw-score* MIAs — those that threshold an absolute membership score (LOSS [3], ZLib [4], Min-K [5], Min-K++ [6]) — because the reference’s artificial easiness suppresses apparent separability. *Reference-normalized* MIAs [7], which subtract a per-example reference-model score, are unaffected, because the offset is common-mode and cancels. At α=0.7, the

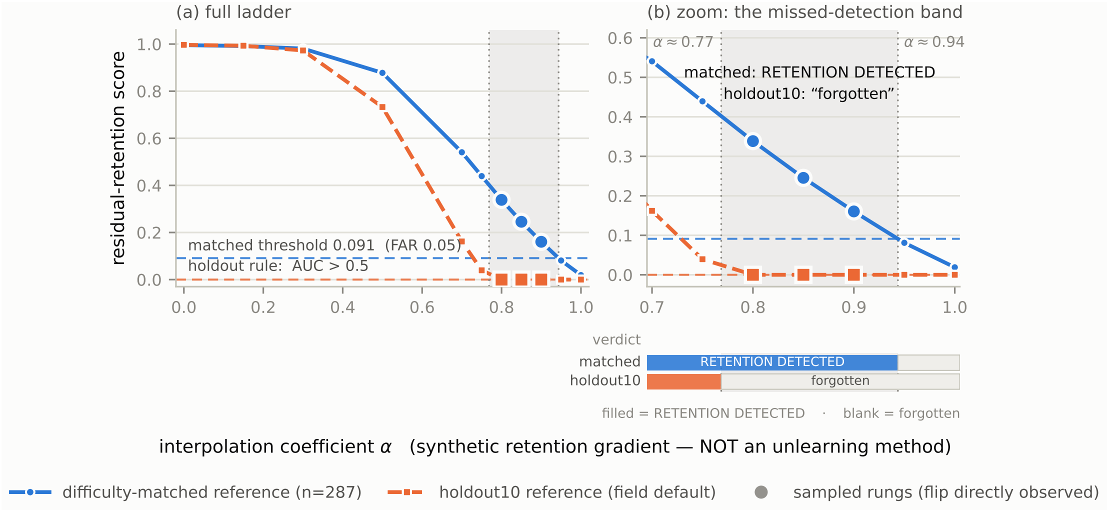

Figure 4: Residual-retention score vs α, under holdout10 (field default) and the difficulty-matched reference, with decision thresholds and the missed-detection band shaded. The verdict strip below panel (b) shows the two references returning opposite verdicts across the band.

residual-retention readout is 0.162 for a raw LOSS attack against holdout10, 0.541 for the same attack against the matched reference, and 0.970 for the reference-normalized attack — a 3.3× under-report by the field-default raw-score/holdout combination.

We note a constraint discovered in constructing the reference-normalized comparison: computing it against the *same* model used as the difficulty null yields an identically-zero score by construction (a self-comparison), so calibrating a reference-normalized MIA requires a null model *distinct* from the reference model. This is a design constraint for any future reference-normalized auditor built on this benchmark.

Table 5: MIA-family map. Raw-score attacks threshold an absolute score against the reference and inherit the offset; the reference-normalised family cancels it per example. The $\alpha = 0.7$ column is the receipt quoted above.

attack family	members	scoring	inherits offset?	ρ at $\alpha = 0.7$
raw score, holdout reference	LOSS, ZLib, Min-K, Min-K++	absolute vs reference	**yes**	0.162
raw score, matched reference	same attacks, matched set	absolute vs reference	mitigated	0.541
reference-normalised	Reference	per-example difference	**no**	0.970

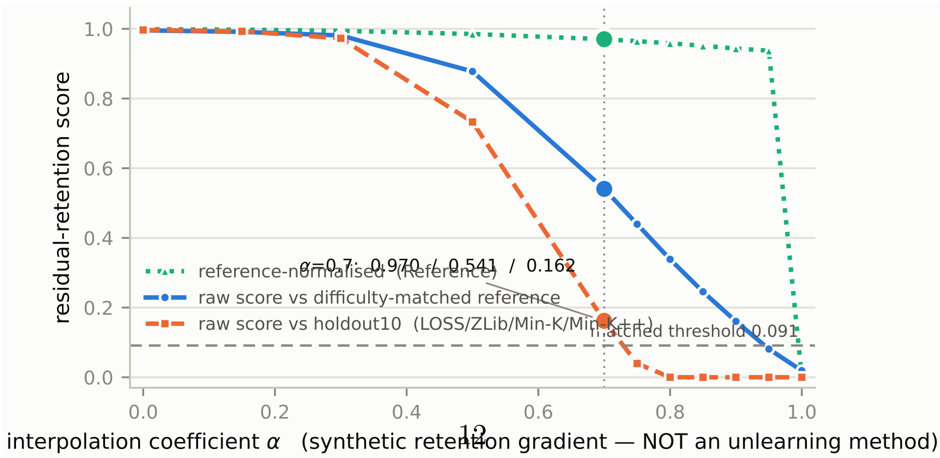

Figure 5: Retention vs α under the three scorings. The raw-score attack against the field-default holdout collapses earliest, the same attack against a difficulty-matched reference degrades later, and the reference-normalised scoring — which cancels the offset per example — holds highest throughout.

Relative rules are immune. Gold-referenced rules (e.g., comparing a candidate to a retrain model on the same non-member set) are robust to the difficulty offset: the offset depresses the candidate's and the gold's scores by the same amount, so subtraction cancels it. Empirically, the reference-set-induced spread collapses from ≈0.20 under absolute rules to ≈0.02 under a relative rule — an order of magnitude.

But the escape hatch is unavailable on TOFU as shipped. A relative rule requires a valid gold retrain to reference. TOFU/Phi provides none: the released retain checkpoint is a byte-identical duplicate of the full finetuned model (identical safetensors weights, differing only in fp32 vs bf16 encoding), and the released unlearned checkpoints (`phi_grad_ascent_*`, `phi_grad_diff_*`, `phi_KL_*`, `phi_idk_*`) are empty repositories. There is no gold retrain to reference without training one. The offset breaks the auditing mode the benchmark supports, and the mode that would survive it is the one the benchmark cannot supply.

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8. A Difficulty-Matched Reference (C5)

We restore calibration by constructing a difficulty-matched non-member reference. For each forget-set example, we select a distinct holdout example whose *never-trained-model* NLL is within a caliper of the forget example's, using only the null model's scores — never the target model's — so no membership signal leaks into the construction. With a caliper of 0.05, this retains 287 of 400 pairs and drives the never-trained separability from AUC 0.332 to 0.509 — matched to noise, the same neighborhood as the original-split control (0.514).

Honest cost. Matching discards the forget set's hard tail, so the matched auditor evaluates a *difficulty-matched subpopulation* of the forget set (n=287/400), not the whole set, with a residual imbalance of +0.015 nats/token that we report alongside every matched result. This qualifier travels with every matched number in the paper.

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9. Discussion

Benchmark construction is an auditing decision. The difference between the two benchmarks we probe is not subject matter, model family, or scale; it is whether the non-member reference was cut from the same cloth as the forget set. MUSE-News partitions one collection pass and passes; TOFU's holdout was regenerated separately, later, and fails. Regeneration looks innocuous — same format, same generator, same topic — and is exactly where ($\star$) silently breaks, since a generator's output distribution is not stable across time, prompt revisions, or model versions. The rule follows directly: **partition one corpus; do not regenerate a reference later.** Where regeneration is unavoidable, the offset is measurable before any auditing is done, at the cost of one forward pass per example with an off-the-shelf model.

Why this matters for the right to erasure. Unlearning is motivated by data-protection regimes in which a subject may demand removal of their data, and an audit is what converts "we ran an unlearning method" into "the data is gone." A miscalibrated reference undermines that conversion in the direction that matters: the offset is *conservative* for false alarms — it makes members look less member-like, so it cannot manufacture a spurious accusation of retention — but an auditor's purpose is to *catch* retention, and the same conservatism becomes a missed detection, certifying clean a model that has demonstrably retained the data. In a compliance setting that asymmetry protects the wrong party.

Threat model. This is not an adversarial finding — nobody built TOFU's holdout to defeat auditing, and we found the offset with a probe that assumes no adversary. But the consequence is exploitable: a provider free to propose the reference their own unlearning is audited against has a lever that requires no tampering with the model, leaving its weights, outputs, and training logs above suspicion. The reference set should therefore be fixed by the auditor or the benchmark, never by the audited party, and validated rather than assumed — the same discipline applied to a control group in any other experimental science.

What a practitioner should do. Before trusting an MIA-based unlearning number: (i) score the forget and reference sets with a model that saw neither and check the AUC is indistinguishable from 0.5, localising the defect before any auditing is attempted; (ii) prefer reference-normalised attacks, which cancel a per-example difficulty offset structurally rather than relying on the sets being matched; (iii) where a graded verdict is needed, verify the calibration null has non-zero spread — a degenerate null means the reference is doing no work, and a threshold derived from it is worse than none. None of this requires training, gradients, or access to the audited model's internals.

How far the rule generalises. The mechanism — an unchecked difficulty mismatch between the forget set and its reference — is the standard hazard of an unmatched control group, and is not specific to language models, NLL, or unlearning: it appears wherever a membership decision compares two sets assumed, rather than shown, to be comparable. What is specific here is that the check is unusually cheap, since a never-trained model of the same family is an off-the-shelf difficulty instrument. We do not claim reference mismatch is widespread — we tested two benchmarks and one passed. We claim the weaker, better-supported thing: matching is required, not automatic, depends on construction, and should be verified rather than presumed.

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10. Limitations

Synthetic retention gradient. The missed-detection demonstration (§6.2) uses weight-space interpolation, not real unlearning methods. The *reference mismatch itself* (§5) is measured on real data with a real, never-trained probe; the gradient only supplies a graded axis with known ground truth to show the mismatch has consequences under controlled conditions. Validating the missed-detection result on real unlearning methods (grad-ascent, NPO [12], etc.) requires training those methods and a gold retrain — deferred to future work, and the natural extension of this paper.

The offset is TOFU-specific; what generalizes is the check, not the defect. We tested a second benchmark and it passed (§5.1), so we do not claim reference mismatch is widespread. We claim the weaker and better-evidenced thing: a difficulty-matched reference is required but not automatic, it depends on construction, and it is cheap to verify. Two benchmarks is still two — whether WMDP [13] or other references pass is untested, and one clean benchmark is not evidence that most are clean.

One probe-model family. Both benchmarks were probed with base Phi-1.5. A difficulty measure is model-dependent in principle, so a second probe family (Pythia, GPT-2-scale, a larger base model) would strengthen the claim that these are properties of the *data* rather than of one tokenizer and one pretraining mix. The controls make this unlikely to be an artifact — under the same probe, TOFU's two original-generation splits are matched (0.514) while its regenerated holdout is not (0.332) — but it remains untested.

Prose versus QA. The two benchmarks also differ in format: TOFU is short QA scored over the answer span, MUSE-News is long-form prose scored over a fixed 512-token prefix. That the probe behaves sensibly on both is reassuring, but format and construction are not fully disentangled by two data points.

Matched subpopulation. The restored-calibration results audit a difficulty-matched subpopulation of the forget set, not its entirety.

Difficulty proxy. The never-trained model provides a difficulty proxy and a null substitute, not a true gold retrain; a trained gold retrain would sharpen both the matching and the relative-rule analysis.

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11. Conclusion

Membership-inference auditing of unlearning is only as trustworthy as its non-member reference, and that reference is almost never checked. On the field's most-used LLM unlearning benchmark, the standard reference carries a measurable difficulty offset that renders absolute-threshold auditing uncalibratable and induces missed detections, corrupting the raw-score MIAs the field ships. The offset-robust alternative is unavailable on the benchmark as delivered. Auditing research should validate the *reference*, not only the estimator: an audit calibrated against a warped yardstick reports a warped result, however sophisticated the attack.

Reproducibility

All experiments and result arrays are released. Every table and figure re-derives from the committed arrays without a GPU (verified): `exp1_probe_nll.npz`, `exp2_ladder.npz`, `exp2b_fine.npz`, `exp3_muse_news_nll.npz`, `exp3_muse_books_nll.npz`, with rerun instructions in the experiments README.

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Venues are stated only where confirmed; everything else is cited as an arXiv preprint rather than asserting a publication venue.

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